MATH 381 Exam 2

December 6, 2024

Name: Solutions

Student #: _____

Problem:	1	2	3	Total
Points:	9	8	8	25

INSTRUCTIONS:

- You have 50 minutes to take the test.
- Write your solution below the problem. There is scratch paper at the back of the test.
- If you need extra space to write your solution, use the paper at the back and indicate on which page your solution continues.
- The test is double-sided. Make sure you are reading the backs of pages!
- Unless otherwise stated, show all your work for full credit.
- You are allowed to use one 8.5"×11" sheet of notes, front and back.
- You can use a scientific calculator. Graphing calculators and computers are not allowed.

Good luck!

1. Each day, a person either eats dinner at home or goes to a restaurant.

   - If they eat at home one day, the probability they eat at home the next day is 70% and the probability they eat at a restaurant the next day is 30%.
   - If they eat at a restaurant one day, the probability they eat at home the next day is 90% and the probability they eat at a restaurant the next day is 10%.

   Assume that the person's dinner behavior can be modeled as a two-state ergodic Markov chain with the probabilities described above. Assume also that this behavior has already happened for a long time, so the Markov chain has reached its stationary distribution.

   (a) (5 points) In the long run, what fraction of days will the person eat dinner at home?

Transition probability matrix is

$$P = \begin{array}{c|cc} & H & R \\ \hline H & 0.7 & 0.3 \\ R & 0.9 & 0.1 \end{array}$$

Let π be stationary probability

$$\pi(I - P) = 0$$

$$(\pi_1 \ \pi_2)\begin{pmatrix} 0.3 & -0.3 \\ -0.9 & 0.9 \end{pmatrix} = 0 \qquad \pi_1 + \pi_2 = 1$$

$$0.3\pi_1 - 0.9\pi_2 = 0$$
$$\pi_1 + \pi_2 = 1$$

$$1.2\pi_1 = 0.9$$

$$\boxed{\pi_1 = \frac{3}{4}} \qquad \pi_2 = \frac{1}{4}$$

(b) (2 points) Suppose that eating dinner at home costs \$10 and eating dinner at a restaurant costs \$30. On average, how much does the person spend per day on dinner?

$$\frac{3}{4}(\$10) + \frac{1}{4}(\$30) = \boxed{\$15}$$

(c) (2 points) On any given two consecutive days, what is the probability that the person eats dinner at home both days? (If you are stuck on this, please skip ahead and don't waste time.)

$$P(X_{t+1}=H, X_t=H) = P(X_t=H)\,P(X_{t+1}=H \mid X_t=H)$$

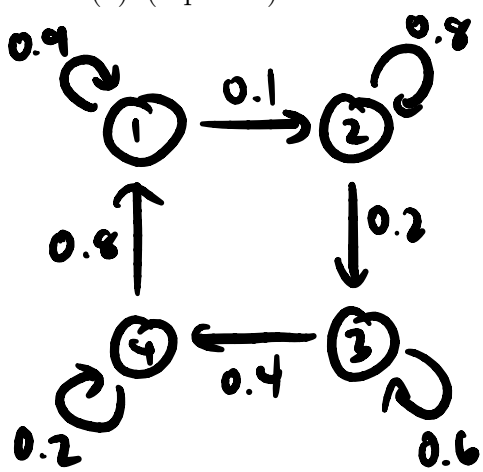
$$= \underbrace{\frac{3}{4}}_{\text{From (a)}} \cdot \underbrace{0.7}_{\text{From transition probability matrix}}$$

$$= \boxed{\frac{21}{40}}$$

2. Let $X_0, X_1, X_2, \ldots$ be a four-state Markov chain with the following transition probability matrix.

$$\begin{pmatrix} 0.9 & 0.1 & 0 & 0 \\ 0 & 0.8 & 0.2 & 0 \\ 0 & 0 & 0.6 & 0.4 \\ 0.8 & 0 & 0 & 0.2 \end{pmatrix}$$

(a) (3 points) Is this Markov chain ergodic? Explain your answer.



Yes. It is irreducible because any state can be reached from any other state in some number of steps. It is aperiodic because any state can move to itself in one step.

(b) (2 points) One of the two vectors

$$(1, 2, 4, 8) \qquad (8, 4, 2, 1)$$

is a left eigenvector associated to 1 for this transition probability matrix. Which one is it? Explain how you know, **without** solving a system of equations. (You do not need to write out all your arithmetic.)

$$(1\ 2\ 4\ 8)\begin{pmatrix} 0.9 & 0.1 & 0 & 0 \\ 0 & 0.8 & 0.2 & 0 \\ 0 & 0 & 0.6 & 0.4 \\ 0.8 & 0 & 0 & 0.2 \end{pmatrix} \neq (1\ 2\ 4\ 8)$$

$$(8\ 4\ 2\ 1)\begin{pmatrix} 0.9 & 0.1 & 0 & 0 \\ 0 & 0.8 & 0.2 & 0 \\ 0 & 0 & 0.6 & 0.4 \\ 0.8 & 0 & 0 & 0.2 \end{pmatrix} = (8\ 4\ 2\ 1)$$

So $(8\ 4\ 2\ 1)$ is the correct answer.

Answers without matrix multiplication were only accepted if they referenced transition probabilities between <u>different</u> states.

(c) (3 points) Find a stationary distribution for this Markov chain. Is it the only stationary distribution for this chain, and why?

$$\frac{1}{8+4+2+1}\begin{pmatrix} 8 & 4 & 2 & 1 \end{pmatrix} = \boxed{\begin{pmatrix} \frac{8}{15} & \frac{4}{15} & \frac{2}{15} & \frac{1}{15} \end{pmatrix}}$$

It is the only stationary distribution because the Markov chain is irreducible.

3. Suppose we place 10 checkers on a 10×10 checkerboard. Call such a configuration of 10 checkers *good* if no two checkers are adjacent (that is, share a side). Let $\mathcal{X}$ denote the set of all good configurations.

Consider the following Markov chain with state space $\mathcal{X}$. Suppose the chain is currently at state x. Choose one of the 10 checkers in x uniformly at random. Then choose an empty square in x uniformly at random. If it is possible to move the chosen checker to the chosen square while remaining a good configuration, then move the checker there and let the result be the next state. If this is not possible (that is, if the move results in a bad configuration), then let x be the next state. See Figure 1.

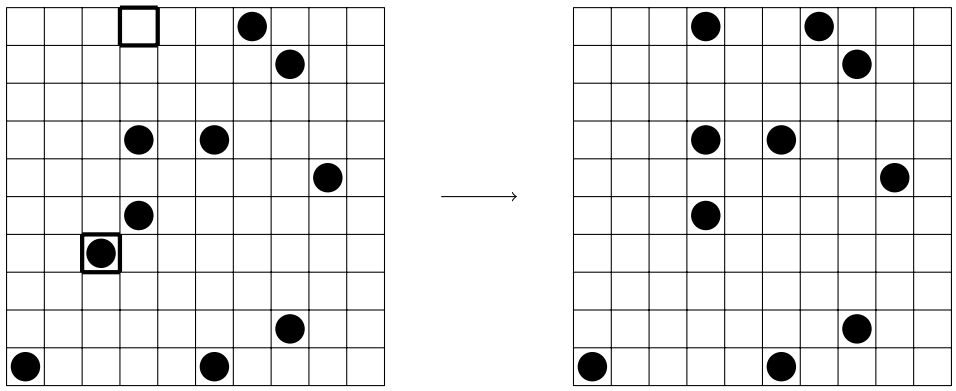


Figure 1: An example transition. The chosen checker and the chosen square are highlighted on the left board.

It can be shown that this Markov chain is ergodic (you don't need to prove this).

(a) (2 points) Suppose that x and y are any two **different** good configurations, and it is possible for the chain to move from x to y in one step. Find the transition probability from x to y. (*Hint:* How many checkers are there, and how many empty squares are there?)

$$\underbrace{\frac{1}{10}}_{\text{Prob. of picking correct checker}} \cdot \underbrace{\frac{1}{90}}_{\text{Prob. of picking correct empty square}} = \boxed{\frac{1}{900}}$$

I also accepted $\frac{1}{910}$.

(b) (2 points) **Briefly** explain why the stationary distribution of this Markov chain is the uniform distribution.

For two different states x and y, either $P_{xy} = P_{yx} = 0$ or $P_{xy} = P_{yx} = \frac{1}{900}$. So P is symmetric, and hence the stationary distribution is uniform.

(c) (3 points) Given a configuration $x \in \mathcal{X}$, let $b(x)$ denote the number of checkers in x which are in the "outer" 36 squares. (An "outer" square is one which touches the edge of the board.) For example, let u and v be the two configurations in Figure 1 (the left one is u). Then $b(u) = 3$ and $b(v) = 4$.

Let $F : \mathcal{X} \to \mathbb{R}$ be the function

$$F(x) = 5^{-b(x)}.$$

We construct a Metropolis chain using the Markov chain on the previous page as the proposal chain, and with stationary distribution proportional to F.

Suppose that we run this Metropolis chain, and at some point, the proposal chain proposes the transition u to v in Figure 1. What is the probability that the Metropolis chain accepts this transition?

$$\alpha = \frac{F(v)}{F(u)} \cdot \frac{P_{vu}}{P_{uv}} = \frac{5^{-b(v)}}{5^{-b(u)}} \cdot 1 = 5^{-4-(-3)} = \frac{1}{5}$$

since P is symmetric

Probability uniformly chosen $r \in [0,1]$ satisfies $r < \frac{1}{5}$ is $\boxed{\frac{1}{5}}$

You needed to state 1/5 was the answer, and not just "the probability that $r < 1/5$".

(d) (1 point) Now suppose that at some point, the proposal chain proposes the transition opposite to the one in Figure 1 (that is, it proposes v to u). What is the probability that the Metropolis chain accepts this transition?

$$\alpha = \frac{F(u)}{F(v)} \cdot \frac{P_{uv}}{P_{vu}} = 5^{-3-(-4)} = 5$$

Probability uniformly chosen $r \in [0,1]$ satisfies $r < 5$ is $\boxed{1}$